

3.4 Hydroxy Compounds

Question Paper

Course	CIEA Level Chemistry
Section	3. Organic Chemistry
Topic	3.4 Hydroxy Compounds
Difficulty	Hard

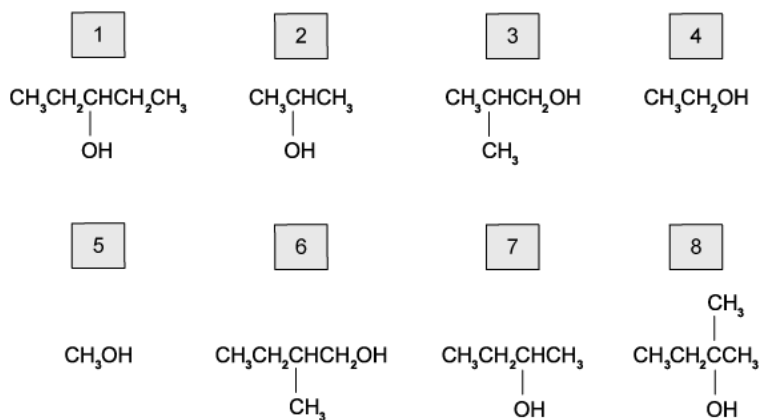
Time allowed: 20

Score: /10

Percentage: /100

Question 1

Which of the following alcohols would give a positive triiodomethane reaction?



- A. 1, 2 and 7
- B. 2, 4 and 7
- C. 3, 4 and 5
- D. 6, 7 and 8

[1 mark]

Question 2

An organic compound **J** reacts with sodium to produce an organic ion with a charge of -3 .

J reacts with NaOH (aq) to produce an organic ion with a charge of -1 .

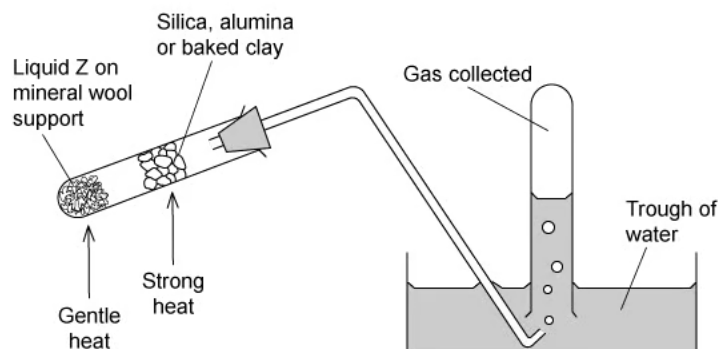
What could be the structural formula of **J**?

- A. $\text{HO}_2\text{CCH(OH)CH}_2\text{CO}_2\text{H}$
- B. $\text{HO}_2\text{CCH(OH)CH}_2\text{CHO}$
- C. $\text{HOCH}_2\text{CH(OH)CH}_2\text{CO}_2\text{H}$
- D. $\text{HOCH}_2\text{COCH}_2\text{CHO}$

[1 mark]

Question 3

The apparatus below shows an organic synthesis reaction:



Which processes could be shown using the above apparatus?

- 1 Dehydration of ethanol (liquid Z)
- 2 Oxidation of ethanol (liquid Z)
- 3 Cracking of paraffin (liquid Z)

A. 1, 2 and 3

B. 1 and 2

C. 1 and 3

D. 1 only

[1 mark]

Question 4

An organic compound **Y** with molecular formula $C_5H_{12}O$, is oxidised to compound **Z** with molecular formula $C_5H_{10}O_2$.

What could be the structural formula of **Y**?

- 1 $CH_3(CH_2)_4OH$
- 2 $CH_3CH_2CH(CH_2OH)CH_3$
- 3 $CH_3C(CH_3)_2CH_2OH$

A. 1, 2 and 3

B. 1 and 3

C. 2 and 3

D. 3 only

[1 mark]

Question 5

A number of alcohols with the formula $C_4H_{10}O$ are separately oxidised. Using 65 g of the alcohols a 55% yield of organic product is achieved.

What mass of product could be obtained?

- 1 42.6 g of butanoic acid
- 2 42.6 g of 2-methylpropanoic acid
- 3 34.6 g of butanone

A. 1, 2 and 3

B. 1 and 2

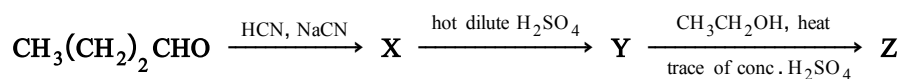
C. 2 and 3

D. 1 only

[1 mark]

Question 6

In a multi-step synthesis reaction, butanal is converted into a compound Z.



What could **Z** be?

- A. $CH_3(CH_2)_3COOCH_2CH_3$
- B. $CH_3(CH_2)_2CH(OH)COOCH_2CH_3$
- C. $CH_3(CH_2)CH(OH)COCH_2CH_3$
- D. $CH_3(CH_2)_2CH(OCH_2CH_3)COOH$

[1 mark]

Question 7

What will react differently with the two isomeric alcohols, $(\text{CH}_3)_3\text{CCH}_2\text{OH}$ and $(\text{CH}_3)_2\text{CHCH}_2\text{CH}_2\text{OH}$?

- A. Sodium
- B. Acidified aqueous potassium dichromate (VI)
- C. Phosphorus pentachloride
- D. Concentrated sulfuric acid

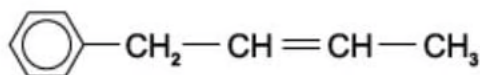
[1 mark]

Question 8

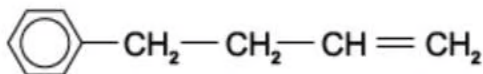
Compound **Q** has the molecular formula $\text{C}_{10}\text{H}_{14}\text{O}$ and is unreactive towards mild oxidising agents.

Which structure formed by the dehydration of compound **Q**?

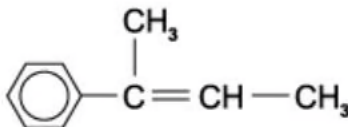
A.



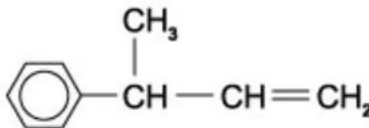
B.



C.



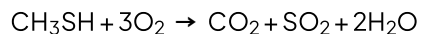
D.



[1 mark]

Question 9

The foul smell that skunks spray is due to a number of thiols, one of which is methanethiol, CH_3SH , which burns as follows.



A sample of 15 cm^3 of methanethiol was exploded with 90 cm^3 of oxygen.

What would be the final volume of the resultant mixture of gases when cooled to room temperature?

- A. 30 cm^3
- B. 45 cm^3
- C. 75 cm^3
- D. 105 cm^3

[1 mark]

Question 10

Use of the Periodic Table is relevant to this question.

1.50 g of butan-2-ol were mixed with excess acidified potassium dichromate (VI). The reaction mixture was then boiled under reflux for fifteen minutes. The organic product was then collected by distillation. The yield of product was 80.0% .

What mass of product was collected?

- A. 0.91 g
- B. 1.17 g
- C. 1.43 g
- D. 1.46 g

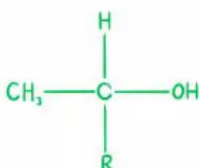
[1 mark]

Model Answers: Hard

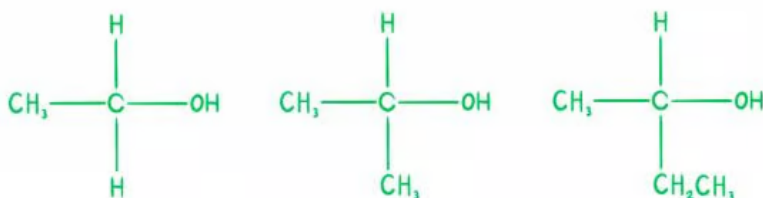
1

The correct answer is **B** because:

- Iodoform, or triiodomethane, tests for the presence of the $\text{CH}_3\text{CH}(\text{OH})-$ group.



- The R can be a hydrogen or an alkyl group.
- If the alcohol contains the $\text{CH}_3\text{CH}(\text{OH})-$ group, then a **pale yellow precipitate** of **triiodomethane**, CHI_3 , is produced.
- Ethanol is the only primary alcohol to give the triiodomethane (iodoform) reaction.
 - Therefore, **compound 4** would give a positive result.
- If R is a hydrocarbon group, the compound will be a secondary alcohol.
- Only secondary alcohols that have a **methyl group** attached to the carbon with the $-\text{OH}$ group will undergo this reaction.
 - Therefore, **compounds 2 and 7** would give a positive result.



A, C & D are incorrect as

- Compounds 3, 5 & 6** are primary alcohols. Ethanol is the only primary alcohol which would give a positive triiodomethane reaction, as R can either be a hydrogen

or an alkyl group.

- **Compound 1** is a secondary alcohol, however, the -OH is attached to a carbon which has an ethyl group on each side, so it would not give a positive triiodomethane reaction.
- **Compound 8** is a tertiary alcohol. No tertiary alcohols can contain the $\text{CH}_3\text{CH(OH)-}$ group because they cannot have a hydrogen atom attached to the carbon with -OH
- Therefore, compound 8 would not give a positive triiodomethane reaction.

2

The correct answer is **C** because:

- Sodium (Na) can react with both carboxylic acid $\text{-CO}_2\text{H}$ and alcohol -OH groups.
- Sodium hydroxide (NaOH) only reacts with the carboxylic acid $\text{-CO}_2\text{H}$ group.
- Consider option **C**:
 - $\text{HOCH}_2\text{CH(OH)CH}_2\text{CO}_2\text{H}$
- This compound contains 2 alcohol groups and 1 carboxylic acid group.
 - In the reaction with sodium, Na would react with all 3 groups and hence form an organic ion with charge -3 .
 - In the reaction with sodium hydroxide, NaOH would react with the carboxylic acid group only and hence form an organic ion with charge -1 .
- Therefore, option **C** is correct.

A is incorrect as this compound contains 2 carboxylic acid groups and 1 alcohol group. **Na**: reacts with all 3 groups and hence would form an organic ion with charge -3 . **NaOH**: reacts with the 2 carboxylic acid groups only and hence would form an organic ion with charge -2 .

B is incorrect as this compound contains 1 carboxylic acid group, 1 alcohol group and an aldehyde group. **Na**: reacts with the alcohol and carboxylic acid group and hence would form an organic ion with charge -2 . **NaOH**: reacts with the carboxylic acid group and hence would form an organic ion with charge -1 .

D is incorrect as this compound contains 1 alcohol group, 1 aldehyde group and 1 ketone group. **Na**: reacts with the alcohol group only and hence would form an organic

ion with charge -1 . **NaOH**: reacts with the **no** groups and hence would form an organic ion with charge 0 .

3

The correct answer is **C** because:

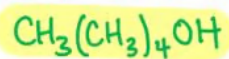
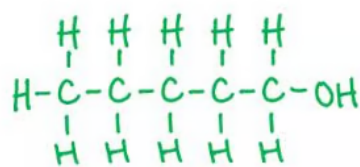
- Ethanol is a **primary alcohol** which can be **dehydrated** to **ethene** gas which can be collected in the test-tube shown in the apparatus above.
 - Dehydration** is the removal of a water molecule from a compound causing the formation of a carbon carbon double bond.
- The reaction takes place when alcohol vapour is passed over a hot catalyst.
- Liquid **paraffin** can also be cracked this way when vapours of paraffin is passed over a heated catalyst to produce a mixture of gaseous short-chain alkane and alkenes.
 - Paraffin is a waxy solid consisting of a mixture of saturated hydrocarbons obtained by distillation of crude oil.

A & B are incorrect as ethanol is a primary alcohol which can be oxidised when heated a solution of potassium dichromate (VI) acidified with dilute sulfuric acid. Primary alcohols will be oxidised to aldehydes and if the aldehyde formed is not distilled off as soon as it is formed, further oxidation to a carboxylic acid will take place. Secondary alcohols will be oxidised to ketones.

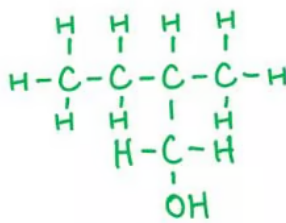
4

The correct answer is **A** because:

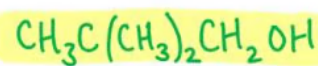
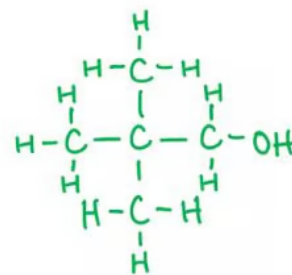
- Since compound **Y** (which contains one oxygen atom) has been oxidised to a compound with two oxygen atoms, compound **Z** must be a **carboxylic acid**.
- Compound **Y** is a primary alcohol which is **oxidised** to a carboxylic acid, compound **Z**, as shown by the two oxygen atoms in its molecular formula.
- All three compounds can be oxidised to a carboxylic acid with molecular formula $C_5H_{10}O_2$.



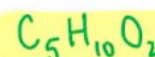
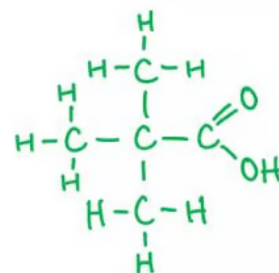
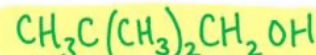
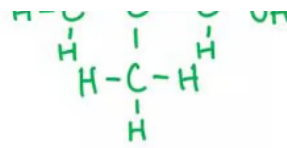
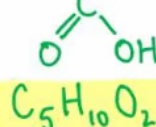
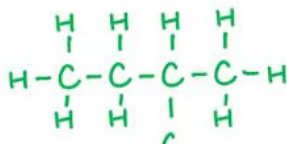
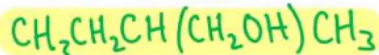
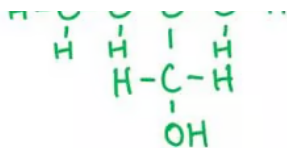
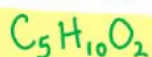
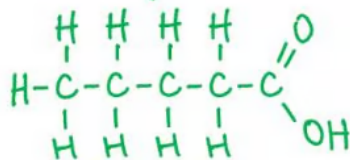
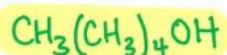
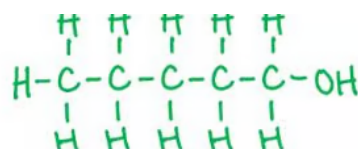
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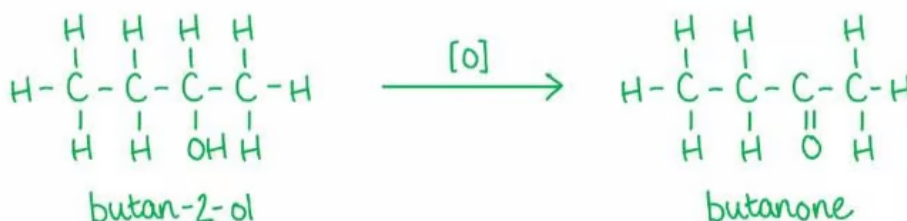
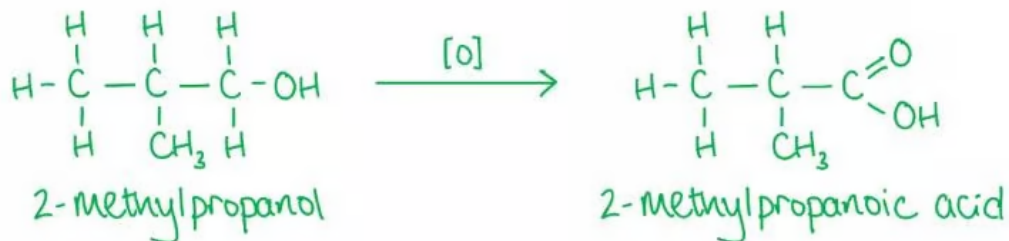
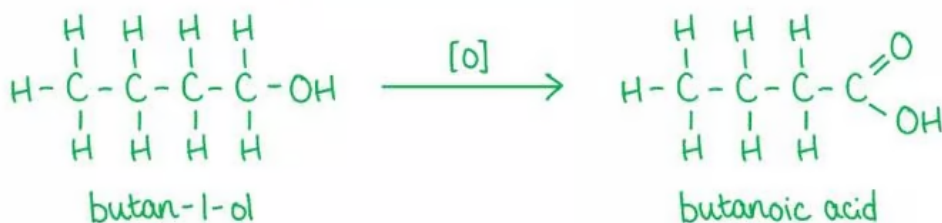


B, C & D are incorrect as all three structural formulae could be compound **Y**.

5

The correct answer is **A** because:

- There can be a mixture of primary, secondary and tertiary alcohols.
 - Only the secondary and primary alcohols can be oxidised.
- Possible structures of alcohols: **butan-1-ol** and **2-methylpropanol** (primary) and **butan-2-ol** (secondary).



- The number of moles of $\text{C}_4\text{H}_{10}\text{O}$ that has reacted is:
 - $\text{moles} = \frac{\text{mass}}{\text{Mr}}$
 - $\text{moles} = \frac{65 \text{ g}}{[(12 \times 4) + (1 \times 10) + 16] \text{ g mol}^{-1}} = 0.88 \text{ moles}$
- Since the yield is 55%, only **0.48 moles** of organic products are formed.
- The masses of organic products can be calculated: **mass = moles x M_r**

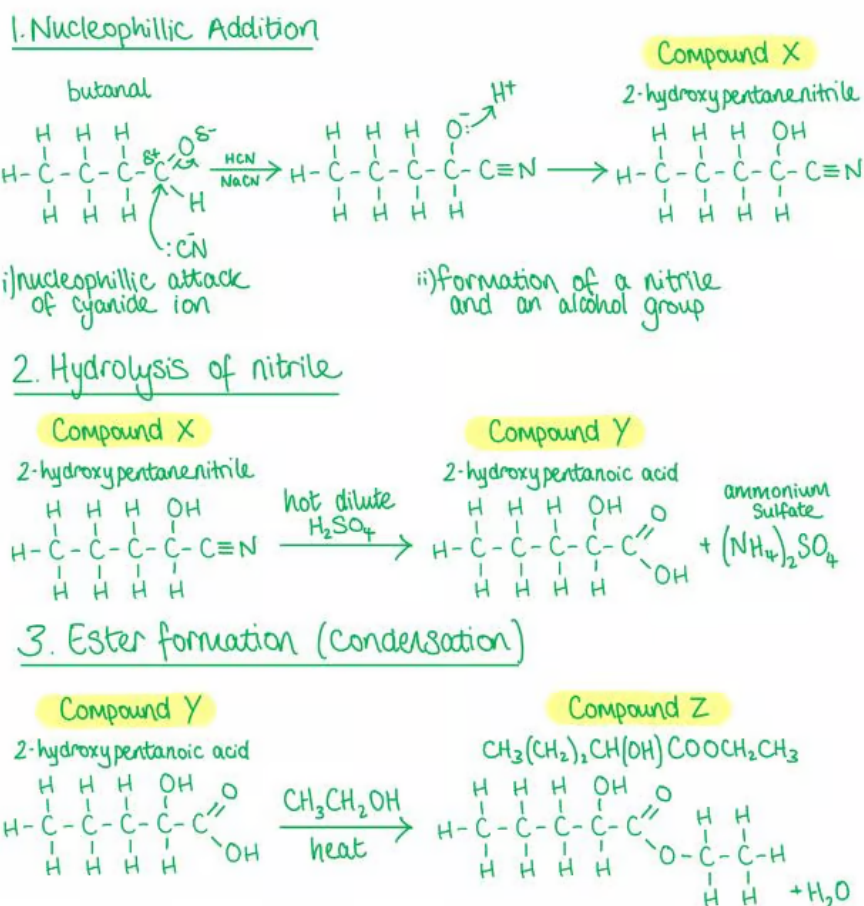
Organic product	Relative molecular mass (g mol^{-1})	Mass (g)
Butanoic acid	88.11 g mol^{-1}	42.6
2-methylpropanoic acid	88.11 g mol^{-1}	42.6
Butanone	72.12 g mol^{-1}	34.6

B, C & D are incorrect as all three masses of product could be obtained.

6

The correct answer is **B** because:

- Butanal is an **aldehyde** which will undergo a **nucleophilic addition** reaction with HCN to form a **hydroxynitrile** (compound **X**).
- Compound **X** is **hydrolysed** by reacting with hot dilute sulfuric acid to form a **carboxylic acid** (compound **Y**) and ammonium sulfate salt.
- Compound **Y** reacts with the alcohol group of ethanol to form an **ester** (compound **Z**).



A is incorrect as the -OH group has been left out.

C is incorrect as the reaction of a carboxylic acid and alcohol gives an **ester**.

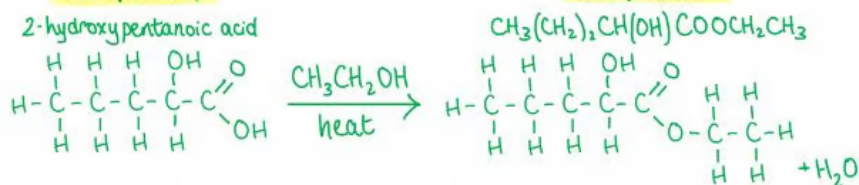
D is incorrect as ethanol reacts at the carboxylic acid group of compound **Y**, not at the

alcohol group.

7

The correct answer is **D** because:

- When an alcohol reacts with concentrated sulfuric acid, dehydration takes place.
 - Dehydration is the loss of a water molecule.
 - This is an **acid catalysed** reaction because H_2SO_4 donates an H^+ which is



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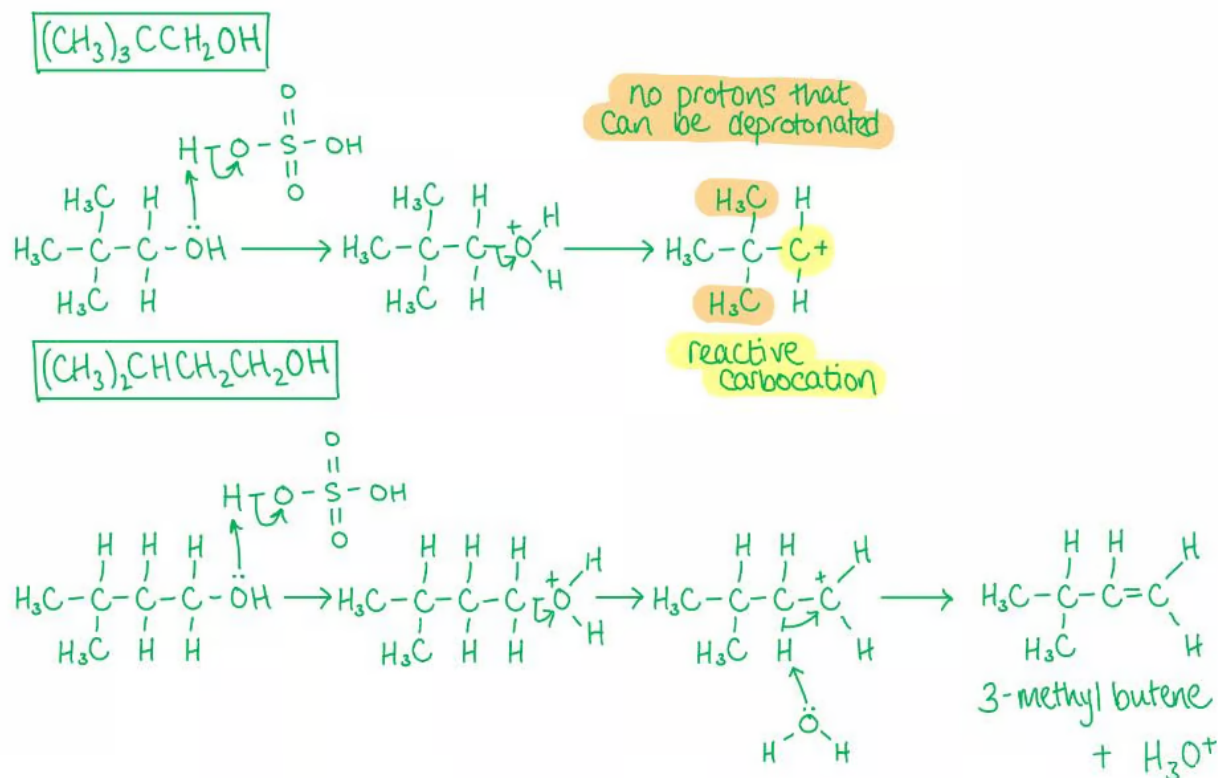
C is incorrect as the reaction of a carboxylic acid and alcohol gives an **ester**.

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The correct answer is **D** because:

- When an alcohol reacts with concentrated sulfuric acid, dehydration takes place.
 - Dehydration is the loss of a water molecule.
 - This is an **acid catalysed** reaction because H_2SO_4 donates an H^+ which is regenerated by the production of H_3O^+ (which dissociates into $\text{H}_2\text{O} + \text{H}^+$).
- For dehydration to take place, the carbon atom **adjacent** to the alcohol must be bonded to hydrogen atoms so can be **deprotonated** (lose a H^+) to form a $\text{C}=\text{C}$ bond – therefore, $(\text{CH}_3)_3\text{CHCH}_2\text{OH}$ **cannot** be dehydrated.



A is incorrect as both alcohols contain an -OH group which will react with reactive metals such as Na to form a salt and hydrogen gas.

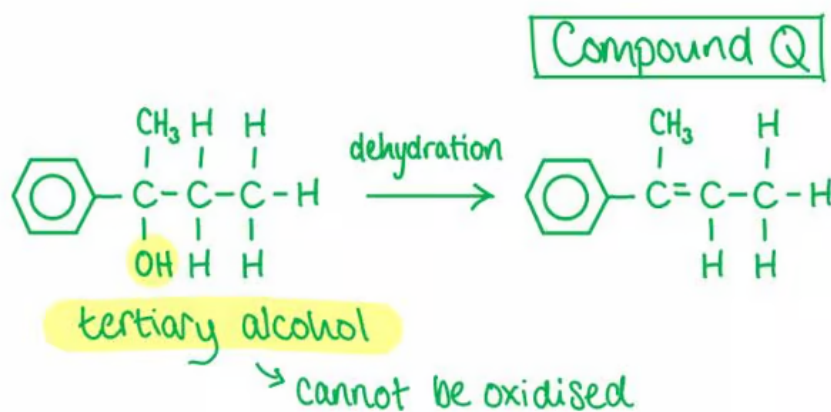
B is incorrect as both alcohols contain a primary alcohol group which will get oxidised to a carboxylic acid.

C is incorrect as both are primary alcohols which will undergo nucleophilic substitution of a chlorine atom to form a primary halogenoalkane.

8

The correct answer is **C** because:

- Both **ketones** and **tertiary alcohols** are unreactive towards mild oxidising agents.
- However, alcohols undergo dehydration to form an alkene, therefore compound **Q** is a **tertiary alcohol**.



A is incorrect as this is the dehydration product of a secondary alcohol which is reactive towards mild oxidising reagents as they get oxidised to ketones.

B & D are incorrect as these are the dehydration product of primary alcohols which are reactive towards mild oxidising reagents as they get oxidised to aldehydes and further oxidised to carboxylic acids.

9

The correct answer is **C** because:

- The question is about the **total volume of gases** when cooled to room temperature.
- At room temperature H_2O is a liquid and not a gas so water can be disregarded.
- Volume of methanethiol combusted: **15 cm^3** .
- An initial, change and final volume table should be made as follows:

	CH_3SH	O_2	CO_2	SO_2
Initial (cm^3)	15	90	0	0
Mole ratio	1	3	1	1
<i>calculation</i>	1×-15	3×-15	$1 \times +15$	$1 \times +15$
Change (cm^3)	-15	-45	+15	+15

<i>calculation</i>	<i>15 – 15</i>	<i>90 – 45</i>	<i>0 + 15</i>	<i>0 + 15</i>
Final volume (cm³)	0	45	15	15

- Since there is 45 cm³ of the initial 90 cm³ O₂ left and 15 cm³ of CO₂ and 15 cm³ of SO₂ produced, the total volume of gases is **75 cm³**.

A is incorrect as the 45 cm³ of leftover O₂ has not been added to the total volume of gases produced.

tertiary alcohol
→ cannot be oxidised

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Change (cm^3)	-15	-45	+15	+15
<i>calculation</i>	$15 - 15$	$90 - 45$	$0 + 15$	$0 + 15$
Final volume (cm^3)	0	45	15	15

- Since there is 45 cm^3 of the initial $90 \text{ cm}^3 \text{ O}_2$ left and 15 cm^3 of CO_2 and 15 cm^3 of SO_2 produced, the total volume of gases is **75 cm^3** .

A is incorrect as the 45 cm^3 of leftover O_2 has not been added to the total volume of gases produced.

B is incorrect as this is the volume of O_2 left.

D is incorrect as the volume of H_2O has also been included in the total volume of gases however at room temperature H_2O is not a gas but a liquid.

10

The correct answer is **B** because:

- Butan-2-ol is a **secondary alcohol** and therefore will oxidise to the **ketone** butanone when refluxed with acidified potassium dichromate (VI).
- **Known quantities:**
 - Mass of butan-2-ol; 1.50 g
 - Molecular weight of butan-2-ol; 74 g/mol *from periodic table*
 - Percentage yield; 80.0%
 - Molecular weight of product (butan-2-one); 72 g/mol *from periodic table*
- **Calculate the moles of butan-2-ol:**
 - $\text{moles} = \frac{\text{mass}}{M_r} = \frac{1.50\text{ g}}{74\text{ g/mol}} = 0.0203$
- **Calculate the yield of product in moles:**
 - $\text{moles} \times \% \text{ yield} = \text{yield of product}$
 - $0.0203 \times 80 / 100 = 0.0162\text{ mol of product}$
- **Calculate the mass of butan-2-one formed:**
 - $\text{mass} = \text{moles} \times M_r = 0.0162\text{ mol} \times 72\text{ g/mol} = 1.166\text{ g} = \mathbf{1.17\text{ g}}$ to 3 significant figures

A is incorrect as this would have been the mass of product if butan-2-ol was dehydrated to butene.

C is incorrect as this would have been the mass of product formed if butan-1-ol had been oxidised to butanoic acid.

D is incorrect as this would have been the mass of the product if there was a 100% yield.